

AMAN SHABILIN BIN SHAMSHULBAHRIN

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SOFTWARE ENGINEER | DATA ENGINEERING & BACKEND

PROFESSIONAL SUMMARY

Software engineer with ~2 years building production data infrastructure after transitioning from chemical engineering. Maintains ETL pipelines and Databricks dashboards processing millions of records; root-caused a 15M-record data coverage gap and resolves 20+ production incidents weekly. Strong in Python, SQL, and cross-functional communication with engineering and business stakeholders.

WORK EXPERIENCE

Data Governance Software Engineer | WeVision Technologies Sdn Bhd

Mar 2026 - Present

Kuala Lumpur, Malaysia

- Maintained consent ETL pipelines and materialized views in Databricks Unity Catalog to ensure governed, reliable dashboard reporting.
- Performed SQL root-cause analysis across source data and transformations, identifying a ~15M-record coverage gap and validating dashboard metrics.
- Redesigned a consent analytics dashboard from 8 static charts into 8 interactive tabs, adding filtering and optimized aggregation for faster reporting.
- Resolved 20+ weekly incidents by monitoring production pipelines in Datadog and Databricks and investigating Unity Catalog anomalies.

Python Developer | AMZ Solutions Sdn Bhd

Oct 2025 - Jan 2026

Kuala Lumpur, Malaysia

- Automated Python ETL pipelines processing daily market and cross-asset risk data, improving consistency and reliability across systems.
- Built RESTful APIs for internal trading analytics and market data, reducing retrieval latency and supporting faster real-time decision-making.
- Partnered with cross-functional teams to translate business requirements into modular, maintainable, and well-documented backend services.

Full Stack Developer | Solution Engineering Sdn Bhd

Jun 2024 - Sep 2025

Kuala Lumpur, Malaysia

- Delivered a backend platform supporting 15 engineering simulation models, enabling real-time system interaction for engineering teams and end users.
- Deployed a Casbin-based RBAC system with SQL-backed policies, improving access control across 30+ organizations and multilayered user roles.
- Designed low-latency backend services that allowed concurrent users to interact with live engineering data without workflow delays.

Programming Intern | Robotics Learning

Jul 2023 - Sep 2023

Kuala Lumpur, Malaysia

- Automated robotic mechanism and control workflows with Python, improving operational efficiency.
- Mentored students in Python and supported improved comprehension, engagement, and test performance.

EDUCATION

Bachelor of Chemical Engineering, Universiti Malaya

Oct 2018 - Jan 2024

PROJECTS

Digital Twin Engineering Simulation

- Engineered a React web-based digital twin platform that visualized live engineering equipment data for real-time monitoring and interaction.
- Developed reusable frontend components for 15 engineering simulation models, reducing development and integration effort.
- Integrated a conversational AI interface that enabled plain-language simulation controls for non-technical stakeholders.

LLM Agent API Backend

- Implemented an AI-powered backend that let engineers query internal documentation directly instead of manually searching across 15 equipment manuals.
- Architected a rate-limited API with per-client throttling and structured error handling to prevent request failures under concurrent AI load.
- Implemented a retrieval-augmented generation (RAG) pipeline to improve response accuracy for domain-specific queries.

TECHNICAL SKILLS

Data Engineering: Python, SQL, ETL/ELT, Data Pipelines, Data Quality, Data Governance, Data Validation

Platforms & Cloud: Databricks, Unity Catalog, AWS S3, DynamoDB

Workflow & Monitoring: Apache Airflow, Datadog, PagerDuty, Production Monitoring

Backend & Tools: PostgreSQL, REST APIs, FastAPI, SQLAlchemy, Git, Docker

Languages: English (fluent), Malay (native)